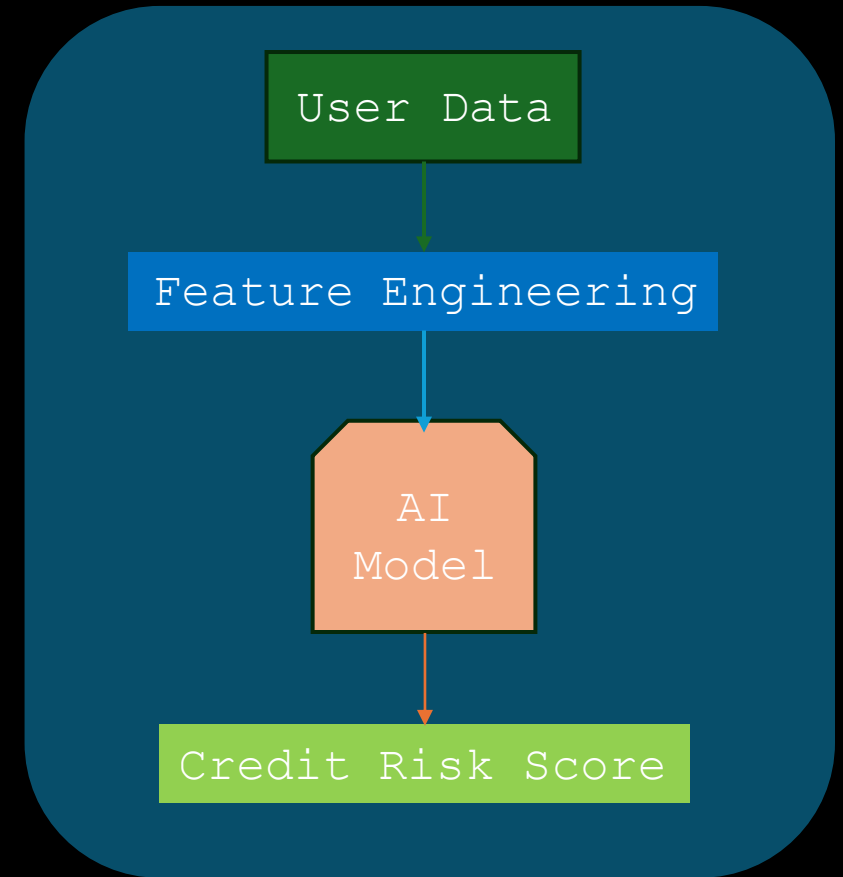


AI Approach: Behavioral & Alternative Credit Signals

- No reliance on traditional credit history
- Uses **behavioral, financial, and stability signals**
- Key inputs:
 - Bill payment consistency
 - Transaction & spending patterns
 - Location stability
 - Bank balance & cash flow
 - **Seasonal income modeling** for informal workers

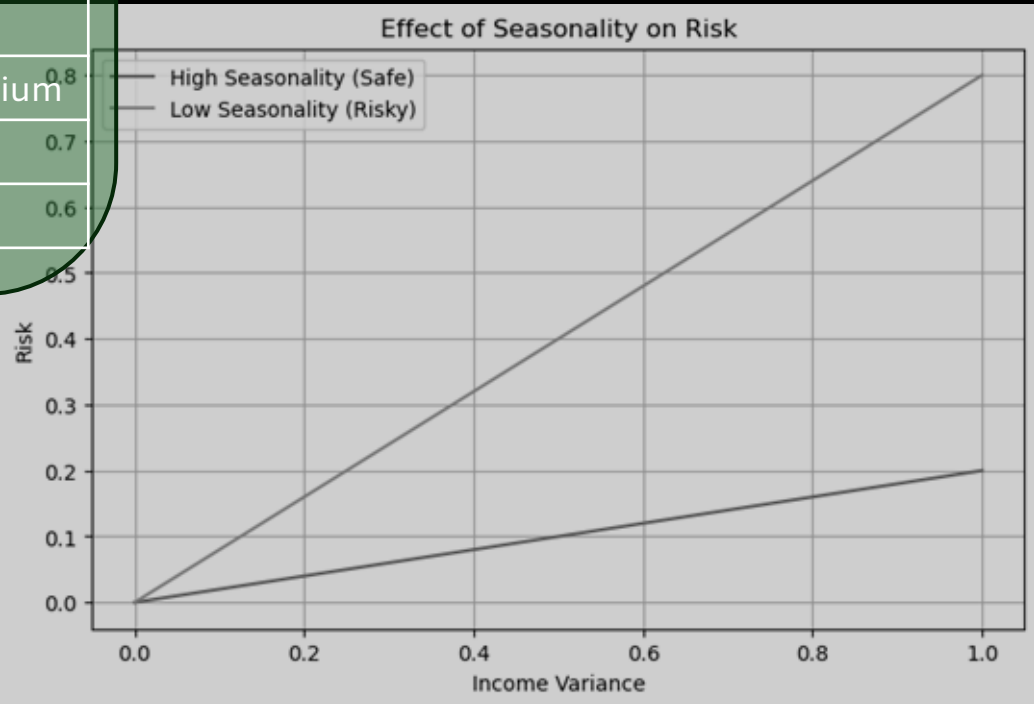
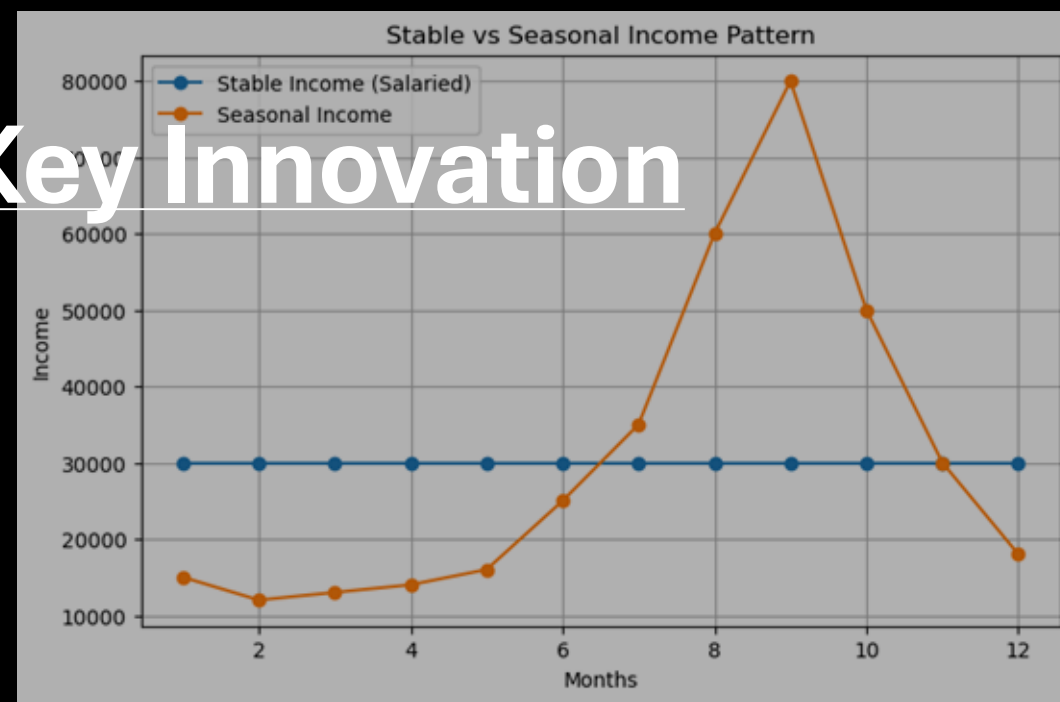


“Creditworthiness is inferred from behavior, not history.”



Feature Engineering & Key Innovation

Feature	Meaning	Importance
bill_payment_score	Repayment discipline	High
ecommerce_spend_ratio	Spending behavior pattern	Medium
location_stability	User traceability & fraud risk	High
balance_norm	Financial strength & liquidity	High
job_stability_score	Income consistency	High
avg_transaction_value	Financial activity level	Medium
transaction_frequency	Banking engagement	Medium
account_age	Long-term financial reliability	Low-Medium
cash_vs_digital_ratio	Financial transparency	Medium
loan_history_score	Past credit behavior	High



Key Innovation:

Seasonality-aware modeling

Differentiates:



random income fluctuations



predictable seasonal spikes



This prevents unfair rejection of seasonal earners like vendors and artisans.

Model Architecture & Training

Models Used:

- Random Forest
- LightGBM
- Logistic Regression

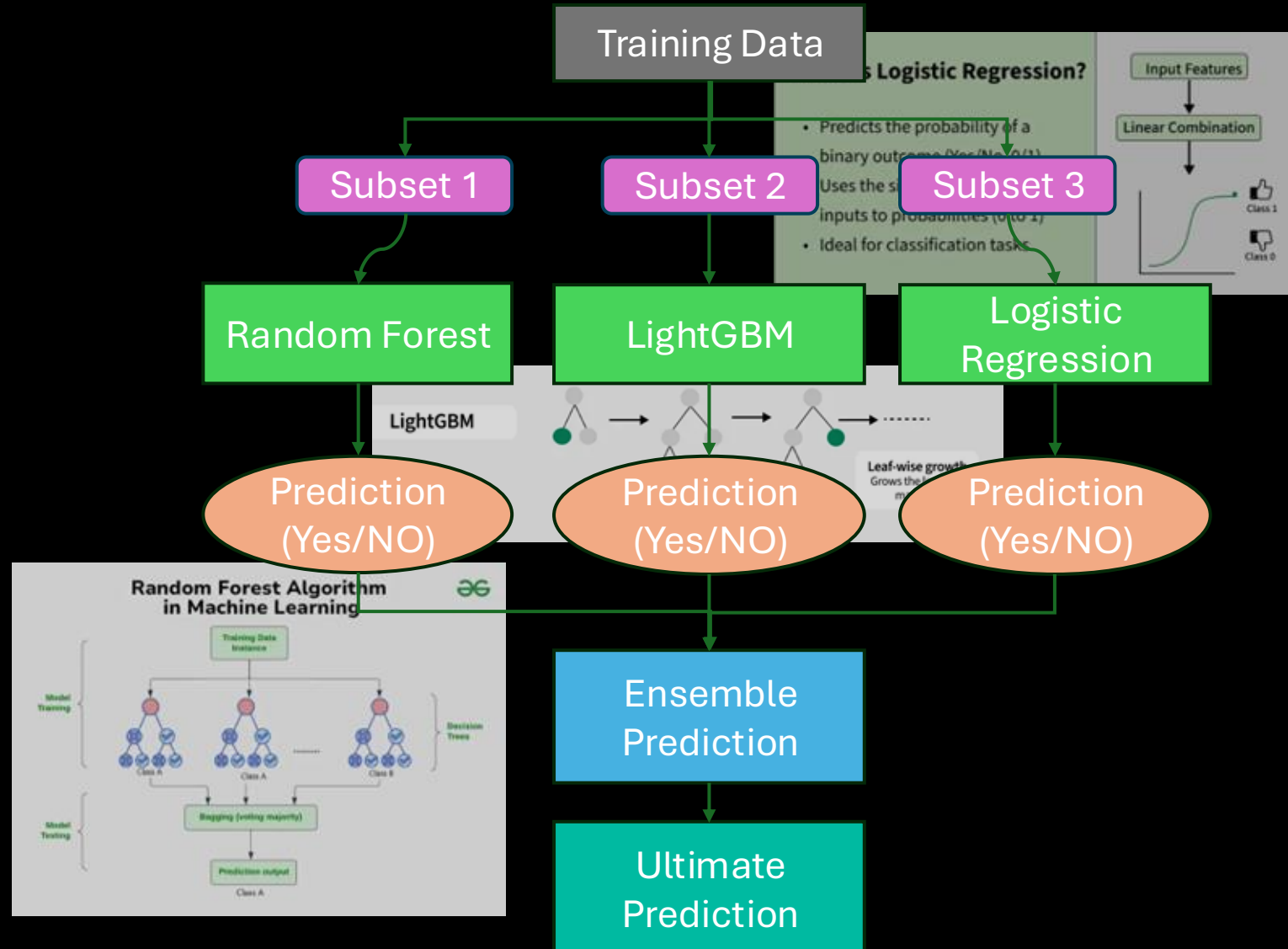
Ensemble Learning (Voting Classifier)

- Combines:
 - accuracy (tree models)
 - interpretability (logistic regression)

Key Design Choices:

- Class imbalance handled (important)
- Focus on minimizing false negatives

Explainable feature importance





Performance & Impact



Model Performance:

- Accuracy: **98%**
- Recall (risky users): **96%**
- High precision & reliability



Key Insights:

- Behavioral features dominate
- Seasonal modeling improves fairness
- Strong performance on imbalanced data



Impact:

- Enables credit access for underserved users
- Reduces NPA risk for banks
- Supports financial inclusion

